

2020 AMC 10A Problem 22 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 22. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 22

Problem:

For how many positive integers $n \leq 1000$ is

$$\left\lfloor \frac{998}{n} \right\rfloor + \left\lfloor \frac{999}{n} \right\rfloor + \left\lfloor \frac{1000}{n} \right\rfloor$$

not divisible by 3? (Recall that $\lfloor x \rfloor$ is the greatest integer less than or equal to x .)

Answer Choices:

- A. 22
- B. 23
- C. 24
- D. 25
- E. 26

Solution:

Clearly, $n = 1$ fails. Except for the special case of $n = 1$,

$$\left\lfloor \frac{1000}{n} \right\rfloor - \left\lfloor \frac{998}{n} \right\rfloor$$

equals either 0 or 1. If it equals 0, this implies that $\left\lfloor \frac{998}{n} \right\rfloor = \left\lfloor \frac{999}{n} \right\rfloor = \left\lfloor \frac{1000}{n} \right\rfloor$, so their sum is clearly a multiple of 3, so this will always fail. If it equals 1, the sum of the three-floor terms is $3 \left\lfloor \frac{999}{n} \right\rfloor \pm 1$, so it is never a multiple of 3. Thus, we are looking for all $n \neq 1$ such that

$$\left\lfloor \frac{1000}{n} \right\rfloor - \left\lfloor \frac{998}{n} \right\rfloor = 1$$

This implies that either

$$\left\lfloor \frac{998}{n} \right\rfloor + 1 = \left\lfloor \frac{999}{n} \right\rfloor$$

or

$$\left\lfloor \frac{999}{n} \right\rfloor + 1 = \left\lfloor \frac{1000}{n} \right\rfloor$$

Let's analyze the first equation of these two. This equation is equivalent to the statement that there is a positive integer a such that

$$\frac{998}{n} < a \leq \frac{999}{n} \implies 998 < an \leq 999 \implies an = 999 \implies a = \frac{999}{n} \implies n \mid 999.$$

Analogously, the second equation implies that

$$n \mid 1000$$

So our only n that satisfy this condition are $n \neq 1$ that divide 999 or 1000. Using the method to find the number of divisors of a number, we see that 999 has 8 divisors and 1000 has 16 divisors. Their only common factor is 1, so there are $8 + 16 - 1 = 23$ positive integers that divide either 999 or 1000.

Since the integer 1 is a special case and does not count, we must subtract this from our 23, so our final answer is $23 - 1 = \boxed{(A) 22}$.

OR

Counting down n from 1000, 999, 998 . . . we notice that $\left\lfloor \frac{998}{n} \right\rfloor + \left\lfloor \frac{999}{n} \right\rfloor + \left\lfloor \frac{1000}{n} \right\rfloor$ is only not divisible by 3 when n is a divisor of only 1000 or 999 (1000, 999, 500, 333 . . .). Notice how the factors of 998 : 1, 2, 499, and 998, do not work.

The prime factorization of 999 is $3^3 \cdot 37$, so $4 \cdot 2 = 8$ factors in total. The prime factorization of 1000 is $2^3 \cdot 5^3$, so $4 \cdot 4 = 16$ factors in total. However, 1 obviously does not work, so we have to subtract 2 (1 is counted twice) from the total.

$$8 + 16 - 2 = \boxed{(A) 22}$$

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